



Linux Fundamentals

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System Logs

System logs on Linux are a set of files that contain information about the system and the activities taking place on it. These logs are important for monitoring and troubleshooting the system, as they can provide insights into system behavior, application activity, and security events. These system logs can be a valuable source of information for identifying potential security weaknesses and vulnerabilities within a Linux system as well. By analyzing the logs on our target systems, we can gain insights into the system's behavior, network activity, and user activity and can use this information to identify any abnormal activity, such as unauthorized logins, attempted attacks, clear text credentials, or unusual file access, which could indicate a potential security breach.

We, as penetration testers, can also use system logs to monitor the effectiveness of our security testing activities. By reviewing the logs after performing security testing, we can determine if our activities triggered any security events, such as intrusion detection alerts or system warnings. This information can help us refine our testing strategies and improve the overall security of the system.

In order to ensure the security of a Linux system, it is important to configure system logs properly. This includes setting the appropriate log levels, configuring log rotation to prevent log files from

becoming too large, and ensuring that the logs are stored securely and protected from unauthorized access. In addition, it is important to regularly review and analyze the logs to identify potential security risks and respond to any security events in a timely manner. There are several different types of system logs on Linux, including:

- Kernel Logs
- System Logs
- Authentication Logs
- Application Logs
- Security Logs

Kernel logs

These logs contain information about the system's kernel, including hardware drivers, system calls, and kernel events. They are stored in the `/var/log/kern.log` file. For example, kernel logs can reveal the presence of vulnerable or outdated drivers that could be targeted by attackers to gain access to the system. They can also provide insights into system crashes, resource limitations, and other events that could lead to a denial of service or other security issues. In addition, kernel logs can help us identify suspicious system calls or other activities that could indicate the presence of malware or other malicious software on the system. By monitoring the `/var/log/kern.log` file, we can detect any unusual behavior and take appropriate action to prevent further damage to the system.

System logs

These logs contain information about system-level events, such as service starts and stops, login attempts, and system reboots. They are stored in the `/var/log/syslog` file. By analyzing login attempts, service starts and stops, and other system-level events, we can detect any possible

access or activities on the system. This can help us identify any vulnerabilities that could be exploited and help us recommend security measures to mitigate these risks. In addition, we can use the `syslog` to identify potential issues that could impact the availability or performance of the system, such as failed service starts or system reboots. Here is an example of how such `syslog` file could look like:

Syslog

```
shellsession
Feb 28 2023 15:00:01 server CRON[2715]: (root) CMD (/usr/local/bin/backup.sh)
Feb 28 2023 15:04:22 server sshd[3010]: Failed password for htb-student from
10.14.15.2 port 50223 ssh2
Feb 28 2023 15:05:02 server kernel: [ 138.303596] ata3.00: exception Emask 0x0 SAct
0x0 SErr 0x0 action 0x6 frozen
Feb 28 2023 15:06:43 server apache2[2904]: 127.0.0.1 - - [28/Feb/2023:15:06:43 +0000]
"GET /index.html HTTP/1.1" 200 13484 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.149 Safari/537.36"
Feb 28 2023 15:07:19 server sshd[3010]: Accepted password for htb-student from
10.14.15.2 port 50223 ssh2
Feb 28 2023 15:09:54 server kernel: [ 367.543975] EXT4-fs (sda1): re-mounted. Opts:
errors=remount-ro
Feb 28 2023 15:12:07 server systemd[1]: Started Clean PHP session files.
```

Authentication logs

These logs contain information about user authentication attempts, including successful and failed attempts. They are stored in the `/var/log/auth.log` file. It is important to note that while the `/var/log/syslog` file may contain similar login information, the `/var/log/auth.log` file

specifically focuses on user authentication attempts, making it a more valuable resource for identifying potential security threats. Therefore, it is essential for penetration testers to review the logs stored in the `/var/log/auth.log` file to ensure that the system is secure and has not been compromised.

Auth.log

```
shellsession
Feb 28 2023 18:15:01 sshd[5678]: Accepted publickey for admin from 10.14.15.2 port
43210 ssh2: RSA SHA256:+KjEzN2cVhIW/5uJpVX9n50B5zVJ92FtCZxVzzcKjw
Feb 28 2023 18:15:03 sudo:    admin : TTY=pts/1 ; PWD=/home/admin ; USER=root ;
COMMAND=/bin/bash
Feb 28 2023 18:15:05 sudo:    admin : TTY=pts/1 ; PWD=/home/admin ; USER=root ;
COMMAND=/usr/bin/apt-get install netcat-traditional
Feb 28 2023 18:15:08 sshd[5678]: Disconnected from 10.14.15.2 port 43210 [preauth]
Feb 28 2023 18:15:12 kernel: [ 778.941871] firewall: unexpected traffic allowed on
port 22
Feb 28 2023 18:15:15 auditd[9876]: Audit daemon started successfully
Feb 28 2023 18:15:18 systemd-logind[1234]: New session 4321 of user admin.
Feb 28 2023 18:15:21 CRON[2345]: pam_unix(cron:session): session opened for user root
by (uid=0)
Feb 28 2023 18:15:24 CRON[2345]: pam_unix(cron:session): session closed for user root
```

In this example, we can see in the first line that a successful public key has been used for authentication for the user `admin`. Additionally, we can see that this user is in the `sudoers` group because he can execute commands using `sudo`. The kernel message indicates that unexpected traffic was allowed on port 22, which could indicate a potential security breach. After

that, we see that a new session was created for user "admin" by `systemd-logind` and that a `cron` session opened and closed for the user `root`.

Application logs

These logs contain information about the activities of specific applications running on the system. They are often stored in their own files, such as `/var/log/apache2/error.log` for the Apache web server or `/var/log/mysql/error.log` for the MySQL database server. These logs are particularly important when we are targeting specific applications, such as web servers or databases, as they can provide insights into how these applications are processing and handling data. By examining these logs, we can identify potential vulnerabilities or misconfigurations. For example, access logs can be used to track requests made to a web server, while audit logs can be used to track changes made to the system or to specific files. These logs can be used to identify unauthorized access attempts, data exfiltration, or other suspicious activity.

Besides, access and audit logs are critical logs that record information about the actions of users and processes on the system. They are crucial for security and compliance purposes, and we can use them to identify potential security issues and attack vectors.

For example, `access logs` keep a record of user and process activity on the system, including login attempts, file accesses, and network connections. `Audit logs` record information about security-relevant events on the system, such as modifications to system configuration files or attempts to modify system files or settings. These logs help track potential attacks and activities or identify security breaches or other issues. An example entry in an access log file can look like the following:

Access Log Entry

shellsession

```
2023-03-07T10:15:23+00:00 servername privileged.sh: htb-student accessed  
/root/hidden/api-keys.txt
```

In this log entry, we can see that the user `htb-student` used the `privileged.sh` script to access the `api-keys.txt` file in the `/root/hidden/` directory. On Linux systems, most common services have default locations for access logs:

Service	Description
Apache	Access logs are stored in the <code>/var/log/apache2/access.log</code> file (or similar, depending on the distribution).
Nginx	Access logs are stored in the <code>/var/log/nginx/access.log</code> file (or similar).
OpenSSH	Access logs are stored in the <code>/var/log/auth.log</code> file on Ubuntu and in <code>/var/log/secure</code> on CentOS/RHEL.
MySQL	Access logs are stored in the <code>/var/log/mysql/mysql.log</code> file.
PostgreSQL	Access logs are stored in the <code>/var/log/postgresql/postgresql-version-main.log</code> file.
Systemd	Access logs are stored in the <code>/var/log/journal/</code> directory.

Security logs

These security logs and their events are often recorded in a variety of log files, depending on the specific security application or tool in use. For example, the Fail2ban application records failed login attempts in the `/var/log/fail2ban.log` file, while the UFW firewall records activity in the `/var/log/ufw.log` file. Other security-related events, such as changes to system files or settings, may be recorded in more general system logs such as `/var/log/syslog` or `/var/log/auth.log`. As penetration testers, we can use log analysis tools and techniques to search for specific events or patterns of activity that may indicate a security issue and use that information to further test the system for vulnerabilities or potential attack vectors.

It is important to be familiar with the default locations for access logs and other log files on Linux systems, as this information can be useful when performing a security assessment or penetration test. By understanding how security-related events are recorded and stored, we can more effectively analyze log data and identify potential security issues.

All these logs can be accessed and analyzed using a variety of tools, including the log file viewers built into most Linux desktop environments, as well as command-line tools such as the `tail`, `grep`, and `sed` commands. Proper analysis of system logs can help identify and troubleshoot system issues, as well as detect security breaches and other events of interest.

Connect to HTB



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